

ECN 594: Introduction and Foundations

Nicholas Vreugdenhil

Plan for today

1. **What is industrial organization?**
2. Course structure and logistics
3. Why do we need demand models?
4. Ordinal vs cardinal utility
5. Random utility framework
6. The logit model
7. Berry (1994) inversion
8. The identification problem

Motivation

- Consider two forms of competition: **perfect competition** and **monopoly**.
- We can think of these two forms of competition as polar opposites.
- **Perfect competition** (supply and demand)
 - Many tiny 'atomistic' firms
 - Total surplus is maximized and so the market is 'efficient'
- **Monopoly**
 - One single firm
 - Monopoly sets price 'too high'; the market is inefficient

Most real-world markets do not fit neatly into these two categories

- Some examples in this course:

1. Firms set different prices for the same good (airline tickets, student discounts)
2. Only a few big firms in the market (health insurance, tech)
3. Firms collude to raise prices (OPEC, cartels)
4. Firms merge with other firms (subject to **antitrust** laws)

What is industrial organization (IO)?

- **IO is the study of firm and consumer behavior in markets between (and including) the polar opposites of perfect competition and monopoly.**
- Why is this useful?
 1. Designing regulation/antitrust policy:
 - When do markets fail? How should we regulate?
 2. Firm strategy:
 - How to set prices? Design marketplaces?

IO is central to many policy debates right now

- Should we break up Google, Amazon, Meta?
- Are hospital mergers raising healthcare costs?
- Can algorithms learn to collude?

These are all IO questions.

This course has two components

- **Theoretical:** Models of pricing, oligopoly, entry, mergers, vertical relationships
- **Empirical:** Demand estimation, merger simulation
- The empirical content (demand estimation) is typically taught at the PhD level
- But it's so useful in practice that we cover it here
- You'll estimate demand using Python and the `pyblp` package

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Course structure

1. **Part 1 (Lectures 1-7):** Demand Estimation and Pricing

- Random utility models, logit demand
- Identification, instrumental variables
- Estimation with `pyblp`
- Price discrimination

2. **Part 2 (Lectures 8-14):** Competition and Industry Structure

- Cournot, Bertrand, Hotelling
- Entry, mergers, collusion
- Vertical relationships

Logistics

- E-mail: nvreugde@asu.edu
- Office: CRTVC 455G
- Office Hours: By appointment (email me)
- **Textbook:** Cabral, *Introduction to Industrial Organization* (2nd ed.)
- For demand estimation: selected readings (posted on Canvas).

Assessment

- **20%** Homework 1
- **20%** Homework 2
- **30%** Midterm (Feb 9)
- **30%** Final (Mar 4)
- Exams: 80 minutes, calculator + one two-sided cheat sheet allowed

Building on your prior courses

- From **ECN 565** (Alvin Murphy): Discrete choice econometrics
 - You know logit, probit, MLE
 - We'll apply this to IO problems
- From **ECN 532** (Hector Chade): Game theory
 - You know Nash equilibrium, Cournot, Bertrand, repeated games
 - We'll do quick refreshers and focus on IO applications

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Why is estimating demand useful?

- You may be surprised why we are starting with demand estimation, given that I motivated IO as about firm behavior
- But it turns out that **estimating demand is central to many IO questions**, including:
 - Setting optimal prices
 - Effects of a merger on prices
 - Value of new goods
 - Any question about consumer welfare

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Ordinal vs cardinal utility

- **Ordinal utility:** only rankings matter, not magnitudes
 - $U(A) > U(B)$ means I prefer A to B
 - But " $U(A) = 2 \times U(B)$ " has no meaning
- This is the standard microeconomics assumption
- **Problem:** We want to measure consumer surplus in \$!
 - "How much better off are consumers from this policy?"
 - Need utility to have a meaningful *scale*

Quasi-linear utility makes utility cardinal

- **Solution:** Assume quasi-linear utility

$$U = u(\text{goods}) + y$$

where y is income (or “money left over”)

- Take the derivative with respect to income:

$$\frac{\partial U}{\partial y} = 1$$

- The **marginal utility of income is constant** (and equals 1)
- This means we can measure utility in dollars!

Why this matters

- With quasi-linear utility:
 - $1 \text{ util} = \$1$
 - Consumer surplus has a meaningful interpretation
 - We can add up utility across consumers
- This is the standard assumption in IO demand estimation
- (In contrast to general equilibrium models where income effects matter)

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Random utility framework (refresher from Alvin's course)

- Consumer i chooses among J products
- Utility of consumer i for product j :

$$u_{ij} = V_{ij} + \varepsilon_{ij}$$

- V_{ij} : deterministic component (observed by econometrician)
- ε_{ij} : random component (taste shock)

Random utility framework

- Consumer i chooses product j if:

$$u_{ij} > u_{ik} \quad \text{for all } k \neq j$$

- Choice probability:

$$P(i \text{ chooses } j) = P(u_{ij} > u_{ik} \text{ for all } k)$$

- Different assumptions on ε_{ij} give different models:
 - Type I Extreme Value $\rightarrow$ **Logit**
 - Normal $\rightarrow$ **Probit**
- You covered this in ECN 565; we'll apply it to IO

Discrete choice in IO

- In IO, we typically write:

$$u_{ijt} = x_{jt}\beta + \alpha p_{jt} + \zeta_{jt} + \varepsilon_{ijt}$$

- i : consumer, j : product, t : market
- x_{jt} : observed product characteristics (size, horsepower, etc.)
- p_{jt} : price
- ζ_{jt} : unobserved (by the econometrician) product quality
- ε_{ijt} : idiosyncratic taste shock (also unobserved)
- $\alpha < 0$: price coefficient (higher prices reduce utility)

What is a “market”?

- We index products by market t . What defines a market?
- **Geography:**
 - City? State? Country?
 - Depends on whether products/prices vary across locations
- **Time:**
 - Week? Month? Quarter? Year?
 - Depends on frequency of price changes, data availability
- **Examples:**
 - Cars: national market, annual (BLP 1995)
 - Cereal: city-quarter (Nevo 2001)
 - Airlines: origin-destination route, quarterly

What is $\tilde{\zeta}_{jt}$: more detail

- $\tilde{\zeta}_{jt}$ = unobserved (by the econometrician) product quality
- Examples (in the car market):

What is $\tilde{\zeta}_{jt}$: more detail

- $\tilde{\zeta}_{jt}$ = unobserved (by the econometrician) product quality
- Examples (in the car market):
 - Brand equity (“I just like Toyota”)
 - Advertising effects
 - Design/style
 - Reputation
- Key insight: firms *observe* $\tilde{\zeta}_{jt}$ when setting prices!
- This creates an endogeneity problem (more on this next lecture)

Why isn't income in the utility function?

- You might expect: $u_{ijt} = x_{jt}\beta + \tilde{\alpha}(y_i - p_{jt}) + \zeta_{jt} + \varepsilon_{ijt}$
- But we write: $u_{ijt} = x_{jt}\beta + \alpha p_{jt} + \zeta_{jt} + \varepsilon_{ijt} \quad (\alpha < 0)$
- **Why?** (think about what you know about discrete choice models from Alvin's course)

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- **Why?** (think about what you know about discrete choice models from Alvin's course)
- With quasi-linear utility, income enters linearly
- When comparing alternatives, income *cancels out*:

$$\begin{aligned} u_{ijt} - u_{ikt} &= [x_{jt}\beta + \tilde{\alpha}(y_i - p_{jt}) + \zeta_{jt} + \varepsilon_{ijt}] \\ &\quad - [x_{kt}\beta + \tilde{\alpha}(y_i - p_{kt}) + \zeta_{kt} + \varepsilon_{ikt}] \\ &= (x_{jt} - x_{kt})\beta + \alpha(p_{jt} - p_{kt}) + (\zeta_{jt} - \zeta_{kt}) + (\varepsilon_{ijt} - \varepsilon_{ikt}) \end{aligned}$$

- Only *differences* matter for choice, and y_i drops out

The outside option

- Important: Consumers can choose **not to buy** any product
- This is the **outside option** (product 0)
- Utility of outside option:

$$u_{i0t} = \varepsilon_{i0t}$$

- We normalize non-idiosyncratic components to 0
- All other utilities are *relative* to the outside option
- Why this matters:
 - If prices rise, consumers can “exit” the market
 - This affects elasticities and market power

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The logit assumption

- Random utility: $u_{ijt} = \delta_{jt} + \varepsilon_{ijt}$ where $\delta_{jt} = x_{jt}\beta + \alpha p_{jt} + \zeta_{jt}$ ($\alpha < 0$)
- What distribution should ε_{ijt} have?
- Assume ε_{ijt} is i.i.d. **Type I Extreme Value** (Gumbel)
- CDF: $F(\varepsilon) = \exp(-\exp(-\varepsilon))$
- Why this assumption?
 - Gives us **closed-form** choice probabilities
 - Computationally tractable

Logit choice probabilities

- With Type I Extreme Value errors, the probability that consumer chooses j :

$$P(\text{choose } j) = \frac{\exp(\delta_{jt})}{\sum_{k=0}^J \exp(\delta_{kt})}$$

- This is the **logit** formula (you've seen this in ECN 565)
- Including the outside option ($\delta_{0t} = 0$):

$$s_{jt} = \frac{\exp(\delta_{jt})}{1 + \sum_{k=1}^J \exp(\delta_{kt})}, \quad s_{0t} = \frac{1}{1 + \sum_{k=1}^J \exp(\delta_{kt})}$$

- Note: $s_{0t} + \sum_{j=1}^J s_{jt} = 1$ (shares sum to 1)

Why logit is useful

- **Closed form:** No numerical integration needed
- **Invertible:** Can go from shares to mean utilities (Berry inversion)
- **Elasticities:** Simple formulas for own- and cross-price elasticities

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Berry (1994) inversion: the key insight

- We observe: market shares s_{jt}
- We want: mean utilities δ_{jt} (to estimate β, α)
- **Problem:** How do we get δ_{jt} from s_{jt} ?
- **Berry's insight:** Take the log of shares!

Berry (1994) inversion

- Start with:

$$s_{jt} = \frac{\exp(\delta_{jt})}{1 + \sum_{k=1}^J \exp(\delta_{kt})}, \quad s_{0t} = \frac{1}{1 + \sum_{k=1}^J \exp(\delta_{kt})}$$

- Take logs:

$$\ln(s_{jt}) = \delta_{jt} - \ln \left(1 + \sum_{k=1}^J \exp(\delta_{kt}) \right)$$

$$\ln(s_{0t}) = -\ln \left(1 + \sum_{k=1}^J \exp(\delta_{kt}) \right)$$

- Subtract:

$$\ln(s_{jt}) - \ln(s_{0t}) = \delta_{jt}$$

Berry (1994) inversion: the estimating equation

- We have: $\ln(s_{jt}) - \ln(s_{0t}) = \delta_{jt}$
- Substitute $\delta_{jt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt}$ (where $\alpha < 0$):

$$\ln(s_{jt}) - \ln(s_{0t}) = x_{jt}\beta + \alpha p_{jt} + \xi_{jt}$$

- This is a **linear regression**!
- LHS: can compute from observed shares
- RHS: product characteristics, price, and an error term

Worked example: Berry inversion

- **Question:**

- Market has 2 products plus outside option. Observed shares:

$$s_{0t} = 0.5, \quad s_1 = 0.3, \quad s_2 = 0.2$$

- Compute the mean utilities δ_1 and δ_2 .

Take 2 minutes to solve this.

Worked example: Berry inversion (solution)

- **Solution:**

- Berry inversion: $\ln(s_{jt}) - \ln(s_{0t}) = \delta_{jt}$

- For product 1:

$$\delta_1 = \ln(0.3) - \ln(0.5) = \ln(0.3/0.5) = \ln(0.6) \approx -0.51$$

- For product 2:

$$\delta_2 = \ln(0.2) - \ln(0.5) = \ln(0.2/0.5) = \ln(0.4) \approx -0.92$$

- Interpretation: Both products have negative mean utility relative to outside option

- Product 1 is “better” than product 2 (higher δ)

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The estimating equation

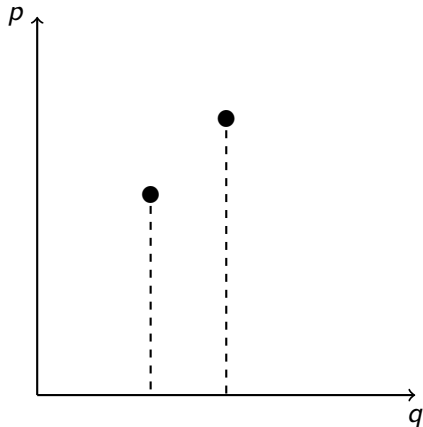
- From Berry inversion:

$$\ln(s_{jt}) - \ln(s_{0t}) = x_{jt}\beta + \alpha p_{jt} + \zeta_{jt} \quad (\alpha < 0)$$

- This looks like a regression we can run with OLS
- **But there's a problem...**

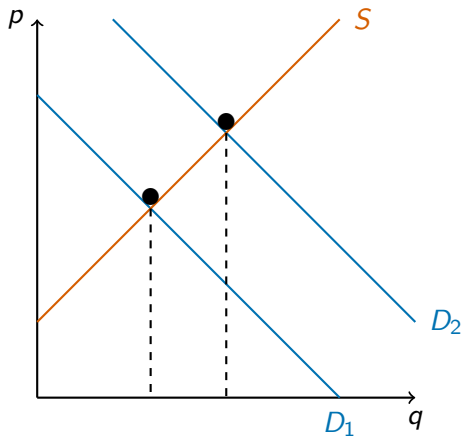
The classic identification problem

- We observe equilibrium prices and quantities
- Question: what's the demand curve?



The classic identification problem

- We observe equilibrium prices and quantities
- Problem: cannot tell if demand or supply is shifting



The classic identification problem

- If we just regress q on p , what do we get?
- Neither demand nor supply!
- **Question:** What variation do we need to identify demand?

The classic identification problem

- If we just regress q on p , what do we get?
- Neither demand nor supply!
- **Question:** What variation do we need to identify demand?
- **Key insight:** We need variation that shifts ONE curve but not the other
 - To identify demand: need **supply shifters**
 - To identify supply: need **demand shifters**

Price endogeneity in demand estimation

- Our estimating equation:

$$\ln(s_{jt}) - \ln(s_{0t}) = x_{jt}\beta + \alpha p_{jt} + \zeta_{jt} \quad (\alpha < 0)$$

- ζ_{jt} = unobserved product quality
- **Problem:** Firms observe ζ_{jt} when setting prices!
- High quality products (ζ_{jt} high) tend to have high prices
- $\Rightarrow \text{Cov}(p_{jt}, \zeta_{jt}) > 0$
- OLS gives biased estimates
- **Next time:** How to solve this with instrumental variables

Key Points

1. **IO** studies firm behavior between perfect competition and monopoly
2. This course has **theory** (pricing, oligopoly) and **empirical** (demand estimation) components
3. **Demand estimation** is central to IO: elasticities $\rightarrow$ market power $\rightarrow$ policy
4. **Random utility:** $u_{ijt} = x_{jt}\beta + \alpha p_{jt} + \zeta_{jt} + \varepsilon_{ijt}$ ($\alpha < 0$)
5. **Berry inversion:** $\ln(s_{jt}) - \ln(s_{0t}) = \delta_{jt}$ turns demand into a regression
6. **Identification problem:** Price is endogenous because firms observe ζ_{jt}